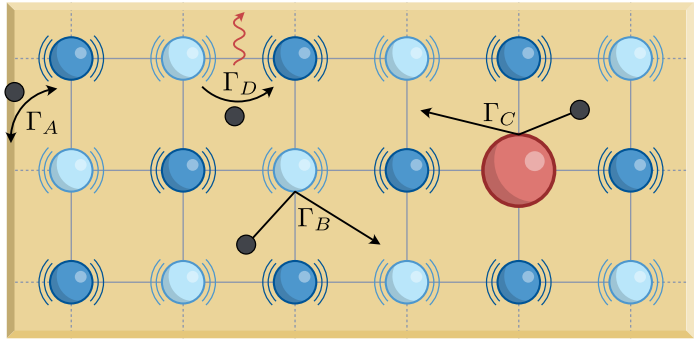




 lattice vibration  disorder  electron  fermionic bath